



Optimizing Wind Energy Integration: A Review of Forecasting Techniques and Emerging Trends

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Received: 16 June 2025 / Revised: 20 September 2025 / Accepted: 12 October 2025

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Abstract

Rapid growth in wind energy highlights the need for accurate forecasting to optimize generation and grid integration. This review analyzes current wind power prediction models, covering their methodologies, strengths, and limitations to guide researchers, engineers, and policymakers. It begins with Numerical Weather Prediction (NWP) models, which are essential yet limited by challenges in complex terrains and localized events. In response, machine learning techniques—such as artificial neural networks, support vector regression, and random forests—have gained prominence for improving forecast accuracy. Advanced methods like bootstrapping and Bayesian model averaging enhance probabilistic forecasts by quantifying uncertainty. The integration of LIDAR and satellite data has further refined wind resource assessment and forecasting accuracy. This review explores the impact of remote sensing and forecasting on decision-making for wind farm operators and grid managers, while also addressing challenges posed by climate change, extreme weather, and the evolution of smart grids. Scenario analysis and grid optimization strategies are discussed, and the review concludes by evaluating current successes and identifying future research needs—emphasizing interdisciplinary collaboration and data sharing. Notably, machine learning improved wind power prediction accuracy by 15% over traditional models. The GFS model achieved an MAE of 0.45 MW and an RMSE of 0.60 MW, demonstrating strong performance in wind energy forecasting.

Keywords Wind power · Deep learning · Prediction of wind energy · SDG 7

Nomenclature

NWP	Numerical Weather Prediction
LIDAR	Light Detection and Ranging
NRA	National Regulatory Authority
ESI	Electronic Supply Industry
AI	Artificial Intelligence
ANN	Artificial Neural Network
WPMS	Wind Power Management System
WPPT	Wind Power Prediction Tool
ARIMA	Autoregressive Integrated Moving Average
ADALINE	Adaptive Linear Neuron
FFNN	Feed Forward Neural Network
MAE	Mean Absolute Error
WASP	Wind Atlas Analysis
RCNN	Recurrent Neural Network
CNN	Convolutional Neural Network
GRU	Gated Recurrent Unit
LSTM	Long Short-Term Memory
RBF	Radial Basis Function
ML	Machine Learning

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SVM	Support Vector Machines
SVR	Support Vector Regression
SF	Sigmoid Function
RF	Random Forest
GB	Gradient Boosting
k-NN	k-Nearest Neighbor
RLNN	Reinforcement Learning with Neural Network
WPD	Wind Power Density
AWS	Airport Weather Station
WB	Weather Buoy
WCI	Wind Chill Index
GA	Genetic Algorithm
TW	Trade Winds
HWS	Horizontal Wind Shear
VWS	Vertical Wind Shear
OW	Oscillating Wind
STF	Short Term Forecast
MTF	Medium Term Forecast
LTF	Long Term Forecast
SFC	Surface Wind
LLW	Low-Level Wind
MLW	Mid-Level Wind

1 Introduction

Sustainable energy balances social, economic, and environmental aspects to meet present energy demands without sacrificing the capacity of future generations to do the same. Climate change, resource depletion, and ecological degradation make the switch from fossil fuels to renewable alternatives vital as the world's energy needs increase. Because it produces clean electricity free of hazardous emissions, wind power is unique among renewable energy sources in terms of its environmental benefits. The abundance and accessibility of Wind, especially in areas with a suitable climate, improve energy security and lessen reliance on imports of fossil fuels. From an economic standpoint, wind energy fosters rural development and creates tasks locally.

Furthermore, it supports decentralized power generation and is appropriate for a range of geographic areas because of its scalability and versatility. Accurate wind forecasting is necessary to realize its full potential to enhance prediction accuracy and maximize the integration of wind energy into power networks. Over the past 20 years, there has been a roughly 70% increase in the world's need for energy, and estimates suggest that this demand continues to rise at a rate of 2.7% annually until 2030. Around 61% of the world's electricity is presently generated by fossil fuels, which release more than 13 Gt of CO₂ a year. On the other hand, 30% of the world's electricity comes from renewable

sources, with wind power alone accounting for over 7% in 2022, or more than 906 GW of installed capacity. China (365 GW), the US (144 GW), and India (43 GW) are the top three nations in terms of wind power deployment. With average levelized prices of \$0.033 per kWh compared to \$0.056 per kWh for coal, onshore Wind has become one of the most affordable energy sources, with costs having dropped by over 56% since 2010. In high-resource areas, modern wind turbines can attain capacity factors of 35% to 50%, which is higher than the solar photovoltaic average of 18% to 25%. Additionally, wind energy sustains more than 1.3 million employment worldwide, promoting energy independence and rural economic prosperity.

In recent times, the electricity demand has witnessed rapid growth, driven by social, economic, and industrial development. However, this surge in demand comes at a time when fossil fuel reserves for power generation are depleting, leading to increased environmental pollution. To ensure optimal resource utilization and provide consumers with cost-effective and quality services, the significance of renewable energy-based technologies is gaining prominence, as they are seen as a primary alternative energy source, and their integration into power systems is rapidly increasing. Power forecasting is essential for operational and scheduling purposes, while price prediction plays a strategic role in safeguarding the financial interests of power generation companies [1–3]. Numerous authors have conducted comprehensive reviews of methodologies and models in various areas, focusing on specific aspects, such as load forecasting and wind power and speed forecasting. Due to the inherent complexity associated with wind power generation, which is influenced by multiple seasonality aspects and natural factors, designing accurate prediction models remains challenging. However, advancements in AI and machine learning have led to the proposal of new models at a rapid pace. In this dynamic environment, it becomes crucial to explore the latest developments in wind forecasting for the benefit of upcoming researchers. The primary objective of this review paper is to conduct an exhaustive review of diverse prediction methods applied to wind power and its related parameters, such as wind speed. The review targets explicitly articles published in prominent international journals after 2015. Various essential aspects are taken into account during the review process, including the model type utilized, the learning algorithm employed, the preprocessing techniques applied, the factors influencing the forecast variable, the prediction time horizon, the criteria for accuracy assessment, the dataset used for analysis, and the NN architecture. The authors have carefully analyzed over 70 research papers, and their observations indicate that neural networks emerge as the predominant approach for estimating wind power and its derivatives. Moreover, hybrid

models have exhibited superior accuracy, indicating that frequent updating of training data during shorter intervals can enhance precision.

The increasing demand for energy in the past few decades has exerted significant strain on current power generation systems, primarily dependent on fossil fuels. This heightened reliance on fossil fuels depletes limited natural resources and exacerbates environmental issues due to the release of carbon dioxide and other contaminants. To tackle these difficulties and move towards a less sustainable power future, countries and states globally are acknowledging the necessity for alternate and environmentally friendly energy sources. Wind energy is a vital element in this shift. Due to the harmful effects and resource constraints of energy from fossil fuels, renewable sources have become widely recognized as a feasible alternative. Wind energy has become a significant contributor to the renewable energy sector. States are diligently launching wind power initiatives, both offshore and onshore, to capitalize on the plentiful and renewable energy supply provided by wind. The scalability and adaptability of wind turbines render them appropriate for diverse sites, from urban centers to isolated locales, facilitating the decentralization of electricity generation. A primary justification for the growing utilization of winds is its capacity to substantially diminish environmental impact. In contrast to conventional fuel-based generation of energy, wind turbines generate electricity without releasing greenhouse emissions, rendering them an essential instrument in combating climate change.

Moreover, wind farms have a comparatively modest physical footprint in relation to conventional power plants, hence reducing habitat disturbance and land use issues. Governments acknowledge the significance of energy safety and autonomy. Dependence on fossil fuel exports for electricity generation presents economic and geopolitical vulnerabilities. Domestically generated wind energy bolsters energy security by diminishing reliance for foreign energy sources. This transition to renewable energy supports the development of a diverse and robust energy portfolio, mitigating susceptibility to supply interruptions and price volatility. In addition to advantages in environmental and energy security, the wind energy sector presents significant commercial prospects. Wind power initiatives invigorate local economies by generating employment in production, installation, and maintenance. Moreover, they can revitalize rural communities by providing a steady income stream for landowners hosting wind turbines. States are recognizing the potential for economic growth and job creation through investments in wind energy infrastructure. In conclusion, the growing need for wind energy in the state's current power generation stems from a confluence of environmental concerns, economic advantages, energy security imperatives, and the

recognition of wind power as a pivotal component of a sustainable and resilient energy future. The proactive integration of wind energy into power systems reflects a collective commitment to reducing environmental impact, enhancing energy security, and promoting economic development at both local and national levels.

Comparing wind power to other renewable energy sources like solar and biomass reveals several benefits. Its energy conversion efficiency is higher, and wind turbines usually run at higher capacity factors, particularly in windy regions [4]. This results in the production of power being more reliable. Moreover, wind farms may coexist with agriculture, reducing habitat disturbance, and require less space than biomass, which requires substantial agricultural fields [5]. Because wind energy can produce electricity during cloudy or colder seasons when solar power is less efficient, it is also more dependable than solar [6]. Economically, wind energy is now among the least expensive energy sources due to its substantial cost reduction. Improvements in energy storage technology improve Wind's grid integration and offset its unpredictable performance [7].

Factors such as data variability, forecasting model complexity, the integration of machine learning approaches, and the impact of temporal features on wind behavior are the leading causes of predicting difficulties in wind power generation. Geographical factors, atmospheric behavior, and turbine-specific traits all contribute to wind power generation variability; hence, data quality is essential for precise forecasting. Operational inefficiencies result from traditional forecasting models' frequent inability to reflect the nonlinear dynamics of wind patterns. Moreover, although machine learning and deep learning techniques enhance accuracy, it must be employed judiciously to avoid overfitting and misinterpretation. The generation of wind energy is significantly affected by temporal considerations, which are often overlooked and can lead to inaccurate predictions. To address these issues, the research introduces sophisticated statistical methodology and hybrid tactics that improve prediction accuracy by integrating the statistical techniques with physical modeling. Emphasizing comprehensive data collection from many geographic regions ensures the acquisition of high-quality datasets, hence enhancing forecasting precision. The study includes a detailed assessment of temporal variables, creating a robust framework for understanding their impact on wind behavior and generating more dependable and accurate predictive models.

Prior studies have achieved notable advancements in wind forecasting over various timeframes. Wang [8] demonstrated the benefits of hybridization by revealing that the integration of ARIMA with artificial intelligence enhanced short-term predictive accuracy by 15%. Dong [9] employed numerical weather prediction for ramp forecasting, resulting

in a 20% reduction in turbine downtime, whereas Jahangiri [10] concentrated on long-term projections utilizing climate models to find areas with high potential. Deterministic models, such as [9], often exhibit suboptimal performance in adverse situations; conversely, ensemble-based techniques, shown by Wilczak and Kim [11, 12], demonstrate greater resilience. Yang and Akpınar [13, 14] also used time-series and Markov models to manage variability, but their practical relevance is limited by their inability to integrate with real-time assimilation. In addition to highlighting the benefits of hybridization and ensemble modeling, these researches highlight the continual challenges in managing uncertainty. In addition to identifying developments like CNNs, LSTMs, and probabilistic ensemble approaches that fill in the same gaps noted in previous publications, this review expands this discussion by methodically contrasting statistical, physical, hybrid, and AI-based models.

In the present research domains that focus on sustainable energy storage infrastructures, it is critical to advance machine learning approaches, especially regarding the difficulties that classic NWP models face. For wind energy to be integrated into the grid as efficiently as possible and to enable the development of more dependable and effective energy storage systems, accurate wind power forecasting is essential. The ability of machine learning techniques like random forests, support vector regression, and artificial neural networks to recognize intricate patterns in past wind data improves forecasting accuracy even in the face of limited topographical influences [15]. These methods make it easier to allocate resources and manage energy, which is essential for creating sustainable energy infrastructures that can adapt to variations in wind production. Additionally, ensemble forecasting techniques like Bayesian model averaging and bootstrapping improve prediction reliability by offering a variety of possible outcomes that can guide risk management and decision-making procedures. By guaranteeing that energy storage solutions are suitably calibrated to satisfy fluctuating energy demands, the integration of these sophisticated forecasting techniques into predictive models directly aids in the development of resilient energy systems and, ultimately, advances the sustainability goals of renewable energy integration and storage. The contribution of this review is as follows:

- **Extensive Discussion of Wind Power Forecasting Models:** This review incorporates all of the main categories of wind power estimating and prediction models (statistical, physical, hybrid, AI/ML-based, ensemble approaches, and data assimilation), in contrast to many previous studies that solely concentrate on one class of models (e.g., statistical, physical, or machine learning).

- **Systematic Comparative Analysis:** With the use of summary tables, this review offers organized comparisons of model types according to their accuracy, strengths, limitations, and guiding principles. This provides a comprehensive approach that is uncommon in previous disjointed evaluations.
- **Highlighting Current Developments:** It demonstrates how contemporary methods outperform conventional models by highlighting recent developments like deep learning, ensemble approaches, and real-time data assimilation.
- **Identification of Research Gaps and Future Directions:** Beyond summarizing existing studies, the review uniquely contributes by highlighting open challenges (e.g., uncertainty, computational efficiency, data limitations) and suggesting future research directions for advancing wind power prediction.

1.1 Questions

1.1.1 Which Artificial Intelligence and Machine Learning Methods are Most Suited to Handle Intricate Spatiotemporal Patterns and Increase the Accuracy of Wind Power Prediction?

The study describes how sophisticated machine learning techniques, such as ensemble approaches and deep learning, can improve forecasting accuracy. It immediately addresses the need for better modeling tools by promoting future research centered on the creation and optimization of these algorithms.

1.1.2 What Techniques May Be Used to Manage and Quantify Uncertainty in Wind Power Predictions?

The study highlights the value of uncertainty quantification strategies in producing probabilistic forecasts and promotes the use of techniques like confidence intervals and probability density functions to lower prediction risks.

The remainder of this review is organized as follows: Section 2 reviews wind prediction models, Section 3 discusses data preprocessing and feature engineering. Section 4 focuses on forecasting models with an estimation technique. Section 5 examines computational efficiency and multi-step forecasting approaches, followed by Section 6, which highlights recent developments, algorithm evaluation, and performance metrics in wind power forecasting techniques. Section 7 discusses key issues and challenges in wind forecasting, and finally, Section 8 concludes the review by outlining research gaps and future directions.

2 Wind Prediction Models

Table 1 provides a comprehensive overview of wind forecasting models, categorized according to their respective time-scales. The period of the horizon refers to the time span considered in wind power forecasting. It is generally classified into short-term, medium-term, and long-term depending on the application requirements, such as grid stability, energy trading, or capacity planning [16, 17]. For every timescale, we delve into detailed descriptions of representative models, including aspects such as consideration of factors, input data utilized, methods employed, and characteristic features.

2.1 Short-Term Prediction

Short-term wind forecasting models frequently employ statistical approaches, with ANN being a preferred choice due to its quicker computation when compared to NWP methods. One notable example of a short-term (immediate) wind prediction model is the WPMS, which has been used for various ICT environments for operators on grids and large wind parks. The WPMS employs an ANN, trained with extensive historical wind farm information. Input data, output data calculated at the wind farm, and the meteorological parameters are preprocessed and converted into an accessible format before being passed into the program [18].

2.1.1 To Summarize, Immediate-Short-Term Forecasting,

- The system calculates the current wind feed for wind power, covering both control zones and specific regions.
- Prediction of wind feed is compiled for the control zone and regions based on prediction-based metrology parameters.

Table 1 Comprehensive overview of wind power forecasting models

Type	Makers	Period of Horizon	Approach Type
Zephyr	INM	Minimum-term	Statistical and physical
WPMS	ISSET, Germany	Minimum-term-Immediate	Statistical
ARMINES (AWPPS)	European Commission	Minimum-term – Immediate-	
ANEMOS	28 partners from 9 countries	Maximum-term – Immediate-	Physical and Statistical
Ewind	AWS Truewind	Minimum-term	Physical and Statistical
WPPT	DTU & IMM	Minimum-term	Physical
Prediktor	IMM	Minimum-term	Statistical

2.2 Short-Range Term Prediction

Numerous tools have been created for short-term wind prediction, including WPFS Ver1.0 and AWPPS. Models have been successfully applied in various case studies across countries like Greece, Ireland, Spain, Denmark, and France. It offers excellent flexibility, providing predictions not only for individual wind farms (e.g., the northern part of Jutland) but also for larger regions (e.g., the Horns Rev offshore wind farm). An essential feature of WPPT is its ability to estimate prediction uncertainties accurately. This is crucial for optimal scheduling and trading decisions. It also incorporates dynamical forecasting models to capture wind power dynamics and potential diurnal variations [19].

Leveraging artificial intelligence, WPPT can adapt automatically to the current conditions, rendering it a valuable predictive tool. In its fundamental arrangement, the system necessitates real-time wind power measurements. Nevertheless, depending on the setup, additional information can be incorporated to augment the precision of the forecasts. Nonetheless, contingent upon the arrangement, the subsequent data is considered:

2.2.1 Real-Time Calculations (measurements) of Wind Power

- Real-time wind power data is collected directly from operating wind turbines.
- Aggregated Power Energy Readings for Regional Forecasting.
- Combines energy readings from multiple turbines within a specific region.
- Enables forecasting of wind power production for larger areas, such as regions or clusters of wind farms.

2.2.2 Speed Prediction of Wind

- Predicted data on the speed of Wind and movement covered wind farms and their respective regions.
- Crucial input for short-range wind power prediction models, allowing for accurate predictions.

2.2.3 Other Types of Measurements or Forecasting

- **Wind Speed (Local):** Additional wind speed measurements from specific locations, aiding in fine-tuning predictions for certain areas.
- **Stability:** Data on atmospheric stability, influencing wind power generation, especially for larger wind farms.

- **Number of Active Turbines:** The number of turbines currently operational impacts the overall generation of power.

Short-term range wind power prediction (or) forecasting utilizes online wind power measurements, aggregated energy readings from various regions of turbines, and meteorological estimation of wind speed.

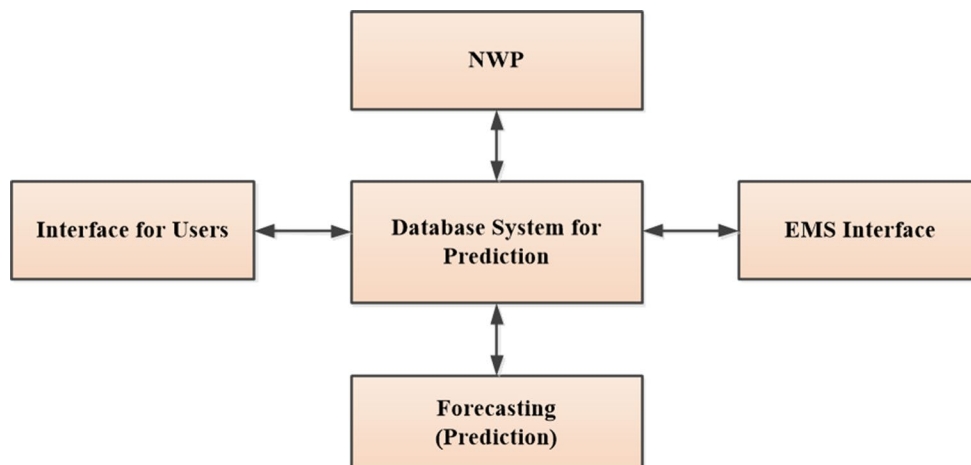
An essential forecasting instrument for the power of Wind is the “predictor,” created by the meteorology research initiative (MET). Diverging from the predictor, which predominantly utilizes physical models to comprehend diverse factors impacting wind power generation. The system offers wind farm production forecasts up to 48 hours in advance, with updates every 6 hours. The local effects, such as roughness, orography (ridges and hills), and obstacles, are then incorporated using the WAsP model [20]. Additionally, park effects, where wind turbines influence each other, are taken into account. To account for any discrepancies between the model predictions and actual observations, Prediktor employs two Model Output Statistics (MOS) filters, which are calibrated using the observed wind power data. It offers forecasts up to 48 or 72 hours in advance, with varying time steps for different time frames. AWPPS stands out for providing confidence intervals for wind power forecasts, allowing users to assess the uncertainty with predefined confidence levels (e.g., 85%, 90%, and 95%).

This model integrates past weather data, wind farm statistics, and generation characteristics, utilizing physical approaches for new wind farms and statistical techniques for existing installations. The system encompasses five primary modules, facilitating enhanced precision and dependability in predictions. As the field of wind power forecasting continues to evolve, researchers and developers are exploring innovative techniques and expanding the capabilities of forecasting tools, as shown in Fig. 1.

2.3 Long-Term Prediction

Modern wind power forecasting tools provide predictions for multiple days ahead and primarily rely on NWP data, making it the preferred option for long-term projections. However, only a few studies have concentrated on long-term wind forecasting methods, resulting in a scarcity of prediction tools for this time frame. Due to the extended forecasting duration, simplistic models are insufficient, and the preference lies with NWP or hybrid NWP models. An exemplar of a long-term wind power prediction tool is “Previento,” which shares similarities with “Predictor” but incorporates more rigorous physical and specialized upscaling techniques [21]. This prediction model optimally amalgamates various weather models while considering local wind farm surroundings and NWP data, guaranteeing accurate and comprehensive forecasts and looking ahead that it is predicted to continue its upward trajectory, as it provides the necessary precision and coverage to address the challenges of forecasting wind power production over extended time periods. It takes into account diverse factors, such as wake effects and orography, while simulating the boundary layer. A crucial aspect in calculating wind speed at hub height involves considering the daily variations of atmospheric thermal stratification, which aids in adjusting the logarithmic profile. By integrating the specific power characteristics of the wind turbine, “Previento” can accurately estimate the anticipated power output for individual sites. Moreover, the system evaluates the total electricity generation from Wind within a designated region by considering selected wind farms. “Previento’s” robust physical modeling allows for dependable forecasts of wind power production at specific sites and regional levels, enabling effective planning and management of wind energy resources.

Fig. 1 Prediction database - wind power prediction



2.4 Hybrid Prediction

ANEMOS represents a versatile wind forecasting tool that caters to diverse time horizons. The project places considerable emphasis on integrating high-quality meteorological forecasting and proper forecasting models, particularly for offshore wind conditions. For offshore forecasting, ANEMOS incorporates marine meteorology and satellite-radar image data, enhancing its accuracy and coverage [22]. The project encompasses both offshore and onshore wind resource prediction, utilizing advanced Statistical, physical, and hybrid approaches to yield improved prediction capabilities.

The project comprises nine work modules, each focusing on specific technical objectives outlined below:

- **Data Collection and Evaluation:** This module deals with the collection and assessment of required data.
- **Online Operation Estimation:** It focuses on estimating online operational parameters.
- **Offline Evaluation of Forecasting Models:** This module involves evaluating forecasting models in an offline setting.
- **Offshore Forecasting:** Dedicated to improving forecasting methods for offshore wind conditions.
- **Development of Physical Methods:** This module concentrates on developing physical forecasting approaches.
- **Development of Statistical Methods:** It is dedicated to creating statistical forecasting methods.
- **Development of ANEMOS Forecasting Platform:** The focus here is on building the ANEMOS forecasting platform.
- **General Dissemination and Assessment:** The final module addresses the dissemination of project outcomes and overall assessment.

This implementation holds the potential to enhance the overall efficiency and dependability of wind generation for

prediction, fostering the sustainable extension of renewable energy sources in the market (See Table 2).

2.4.1 Interval Prediction Modules

- Feature screening, feature learning, intelligent weighted integration, and interval prediction modules are the primary components of the suggested deterministic and probabilistic decomposition ensemble model, which is based on enhanced feature screening and an efficient Gaussian mixed kernel function. To remove the interference of redundant features and capture the best features of the original wind power data, an improved feature screening (IFS) was first suggested in [28]. After that, it was recommended to use an optimal deep neural network to learn the IFS results and investigate the temporal association of these aspects in more detail. The final result was then achieved by integrating the preliminary forecast results using intelligent weighted integration (IWI), taking into account the differences in the contributions of each feature component.

3 Data Pre-Processing and Feature Selection Techniques for Wind Prediction Models

3.1 Data Pre-Processing

When it comes to renewable energy sources, forecasting wind power is crucial for improving the efficiency and balance of wind farms. However, high-quality data used for modeling greatly affects the reliability of wind power forecasts. Ensuring the use of only high-quality data for modeling purposes, data cleaning fixes missing values, irregularities, and conflicts in the dataset. An essential part of pre-processing data is feature engineering. The key is to zero down on the most important parts and tweak them so

Table 2 Data pre-processing techniques of wind prediction models

Technique	Model	References	Variables	Results	Advantages	Disadvantages
Data Normalization	StandardScaler	Häckell, M.W., et al. [23],	Feature values	Scaled data with zero mean	Mitigates scale differences, aids convergence	Sensitive to outliers
Feature Selection	Recursive Feature Elimination	Mir, M., et al. [24],	Feature importance scores	Reduced feature set	Improves model interpretability, reduces overfitting	May discard relevant features
Missing Data Imputation	K-Nearest Neighbors (KNN)	Liu, T., et al. [25],	Features with missing values	Imputed values	Preserves data structure, reduces data loss	computationally expensive
One-Hot Encoding	None	Li, J., et al. [26],	Categorical variables	Binary representation of categories	Works with categorical data, can improve model performance	Increases dimensionality, may lead to multicollinearity
Data Augmentation	Generative Adversarial Networks	Chen, H., et al. [27],	Original data and generated data	Enriched dataset for training	Increases training data diversity, can combat overfitting	Requires significant computational resources for training GANs

they make better predictions. Feature selection strategies help reduce computational complexity and overfitting risk by deleting irrelevant or repeated variables [29, 30].

By ensuring that all values are on a consistent scale, feature transformation techniques like as normalization and standardization prevent specific features from dominating the forecasting procedure owing to their larger magnitudes. It is possible to handle the time or spatial dependencies in wind power data by using approaches that aggregate geographic and temporal data. Data can be more easily analyzed for trends and patterns by aggregating it into important time intervals, such as daily or hourly averages. In order to show a larger geographical area, spatial aggregation brings together data from numerous weather stations or wind farms. Because wind power data often contains complex and nonlinear relationships, it is common practice to employ machine learning methods to resolve these issues. Data normalization or standardization is essential, though, because these procedures may be sensitive to data size and distribution. To avoid biased predictions, resampling techniques like oversampling or undersampling are required for unbalanced dataset management. These datasets fail to adequately represent specific power output levels.

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techniques like oversampling or undersampling are required for unbalanced dataset management. These datasets fail to adequately represent specific power output levels. The model-agnostic One-Hot Encoding method converts category variables to binary ones. It increases dimensionality, but is essential for handling categorical data and improving model performance. Data Augmentation, though not associated with particular references, involves techniques like Generative Adversarial Networks (GANs) to enrich training datasets. It enhances data diversity and combats overfitting, but requires substantial computational resources for GAN training. These techniques collectively form the foundation of adequate data preparation, each with its unique strengths and potential limitations in data analysis and machine learning workflows.

3.2 Feature Selection for Wind Prediction Models

The uncertainty surrounding wind power prediction arises from the unpredictable nature of its derivatives, which can impact the system's reliability [31]. Higher forecast reliability translates to lower operational costs for wind power in the system, leading to potential savings for wind farm owners and improved overall system efficiency.

Figure 2 illustrates periodic time-series curves for price, load, speed, and power of Wind in the Indian electricity market [32]. The load series demonstrates daily periodicity, while the price, wind speed, and power time series exhibit greater volatility and lesser periodicity compared to the load series. Input variable selection involves considering both exogenous and non-exogenous variables.

3.2.1 Physical – Numerical Weather Prediction Models

Most methodologies are based on typical approaches where external factors influence wind, and the forecasting tool obtains input from NWP models. Physical models, being deterministic, encompass the entire array of input derivatives interconnected with wind power generation. Nonetheless, implementing physical models is intricate and time-consuming, as they rely on the properties of wind data collected from metrology stations and the wind farm's location. The provided power curve transforms wind speed into wind power using the equation $W_p = 0.5\rho \cdot B \cdot v^3$, where ρ denotes air density, subject to temperature and pressure, and v represents the Wind through an intercepting area B . This equation takes into account factors, such as the wind turbine and several other variables. NWP models are used to forecast the maximum efficiency of Wind for the surrounding region of the windmill.

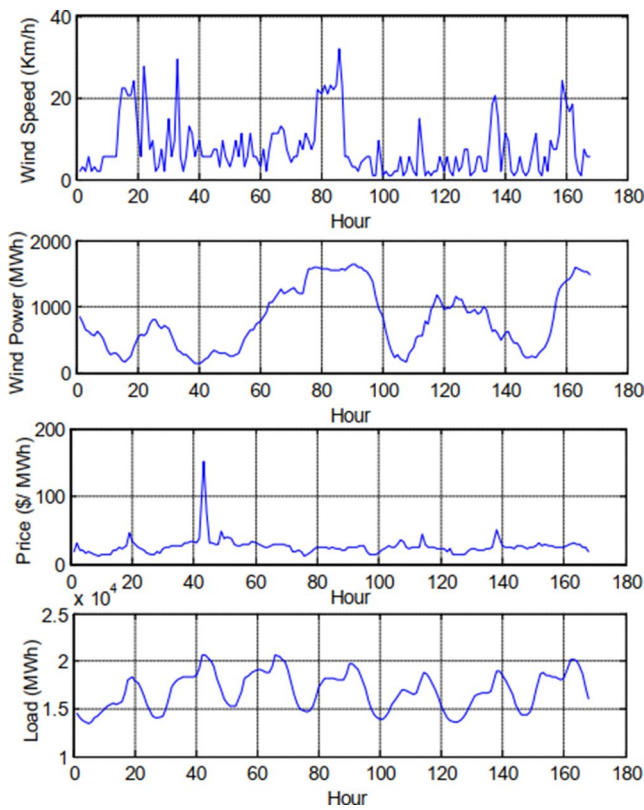


Fig. 2 Periodic time series plots for price, load, speed, and power of wind (January 2019)

3.2.2 Statistical Models

In wind power prediction, various statistical models are utilized for accurate forecasts. ARIMA and SARIMA capture temporal and seasonal patterns, respectively. STL decomposes time series data to analyze seasonal variations [33–35]. GPR offers non-parametric probabilistic predictions with uncertainty estimates. MLR establishes linear relationships between meteorological features and wind power output. GAM extends linear regression for nonlinear interactions.

3.2.3 Hybrid (statistical + Physical (external) Models

The method encompasses the fusion of NWP and statistical techniques for input information selection. NWP data undergoes pre-processing through statistical analysis to identify a suitable time lag for predicting wind forecasting in terms of Hybrid Models. Nonetheless, selecting input variables remains a challenge for power system researchers, given the myriad of factors influencing wind power generation, and 28 variables are shown in Table 3.

Table 3 outlines crucial parameters and variables associated with the input for a specified time period in the context of wind power analysis. These parameters encompass

Table 3 Wind power factors and their variables

Parameters	Variables of Input	Input for Time Period
1. Atmosphere Characteristics	1) Turbulence 2) Radiation 3) Density 4) Rain 6) Humidity	-
2. Topographic Characteristics	7) Turbine Position 8) Elevation 9) Hub Height 10) Degree in Latitude 11) Tower Height	$f(\text{speed of Wind}); (d-m,t), m=1,2,23,7,8,4,3168,365$
3. Wind Power Characteristics	12) Wind power 13) Wind direction 14) Sine and Cosine of Wind 15) Local Wind profile 16) Aggregate wind generation 17) Air density	$f(\text{power of wind}); (d-m,t), m=1,2,4,3,4,67,8,168,365$ and $n=0,1,2,3,4,5,6,7$
4. Behavior Indices	18) Spatial resolution 19) Pressure tendency 20) Temporal behavior 21) Hydrological cycles 22) Radiation – Cloud Interaction	-
5. Geographical Conditions	23) Terrain Roughness 24) Moisture with land surface 25) Snow Covering 26) Mean sea level pressure 27) Obstacles 28) Air temperature	-

various factors, including atmospheric conditions, topographic characteristics, wind power characteristics, behavior indices, geographical conditions, and more. These factors collectively influence the generation and behavior of wind power resources in a given location. Within the “Variables of Input” category, a comprehensive list of variables is provided, ranging from meteorological attributes like turbulence, radiation, density, and humidity to turbine-specific variables such as turbine position, elevation, hub height, and wind direction. Moreover, mathematical representations such as sine and cosine functions of winds, local wind profiles, and cumulative wind generation enhance the comprehension of wind power dynamics. The “Input for Time Period” portion lays out the study’s temporal characteristics, drawing attention to the significance of time intervals $(d-m,t)$ where ‘m’ is a number between 1 and 8 and ‘t’ represents different time periods. The additional collection of time intervals denoted by ‘n’ also ranges from 0 to 7. In order to study the dynamics, variability, and efficiency of wind power generation throughout time, temporal variables are essential. With its extensive set of features, variables, and temporal elements, this table offers a comprehensive framework for researching and simulating wind generation.

This lays the groundwork for conducting in-depth studies of wind and other renewable energy sources, as well as their dynamics, in a variety of climates and locations.

4 Wind Power Forecasting Methods: Estimation Techniques and Approaches

Table 4 delineates the unique models of the neural network procedures, providing a succinct summary of the utilized models, the applied pre-processing approaches, the input data sets employed for training, and the variables that are entered utilized by different researchers. The techniques of Machine Learning & Deep learning are extensively employed methodologies in wind forecasting techniques, with the subsequent five categories of each delineated below.

4.1 Deep Learning Techniques for Wind Power Prediction

Deep Learning techniques have emerged as powerful tools for wind prediction due to their effective model in handling time series data, which exhibit intricate temporal patterns. Traditional statistical models may struggle to capture these intricate dependencies, leading to suboptimal forecasts. In contrast, Deep Learning models excel at capturing long-term dependencies and spatial patterns in sequential data. As a result, the integration of DL techniques in wind power prediction has garnered significant contributions in research and holds immense potential for practical implementation in renewable energy management systems [36]. The significant Deep Learning Techniques are explained below:

4.1.1 Neural Network for Feedforward

Among the many machine learning models used for wind energy predictions is the Feedforward Neural Network (FFNN). While the input layer gathers weather data, the hidden layers do the heavy lifting of analysis and data extraction. We get the wind power prediction from the output layer. Reducing prediction errors is the goal of training the FFNN, which comprises adjusting the connection weights

by backpropagation. Wind power generation predictions are a strong suit of its because of its ability to absorb historical data and extrapolate to new inputs. There are still major challenges with FFNN-based wind turbine forecast models, such as finding the right architecture and avoiding overfitting.

4.1.2 Reinforcement Learning System

A subset of neural networks known as Recurrent Neural Networks (RNNs) excels at predicting future wind speeds and directions. A recurrent neural network (RNN) differs from a feed-forward neural network (FNN) in that it can process sequential data and remember its internal states thanks to its looping architecture. When it comes to wind generation forecasting, RNNs have the potential to accurately capture the temporal connections found in meteorologist time series data, such as wind speed and direction. This makes them especially handy when there are both consistent patterns and dynamic fluctuations in the wind energy production process. To improve knowledge and knowledge retention over long periods of time, the Long Short-Term Memory (LSTM) variant of RNNs counteracts the disappearing gradient problem.

4.1.2.1 Layered Neural Network The impressive performance of Convolutional Neural Networks (CNNs) in wind power forecast approaches illustrates its adaptability beyond their conventional computer vision uses. Since wind patterns often exhibit complex geographical relationships, Convolutional Neural Networks (CNNs) are ideal for wind power forecasting due to their exceptional ability to detect correlations and patterns in spatial data. When it comes to wind power forecasting, convolutional neural networks (CNNs) scour a mountain of historical weather data in search of meaningful patterns and correlations between variables like wind speed, temperature, relative humidity, and more. Using a multitude of convolutional layers, CNNs are able to learn complex nonlinear relationships by independently extracting hierarchical features from input data. The ability to uncover intricate relationships allows for very accurate forecasts of wind power. The impressive

Table 4 Neural networks-based wind power prediction technologies

Model - NN	Algorithm for Learning	Variables for Input	No. of Input Neurons - Total (Time)	Techniques for Pre-Processing
RLNN	DEA	18,19,20	-	MI
RBF	MRAN, GGAP	18,19,20,55	-	WT
RBF	OLS, PSO	20,21,20,55	-	WT, p-ARTMAP
MPL	BP	20	6 Input Series, 31 input	SEA, K-S Test, PDF
RBF	OLS	18,19,20,48,12	13 Neurons	CC, SD
MLP	EPSO	17,18,3,13	22,23,24 Neurons	SOM, Fuzzy
MLP	LM	21,18,2,3,11,44	5 Neurons	9 BCD Clusters, ACF

performance of CNNs in wind power forecasting highlights their importance in improving energy the administration of resources and creating a greener, more efficient energy infrastructure [37].

4.1.3 Recurrent Units with Gates

Managing reliance across time and sequential data is an area where GRUs have shown to be very beneficial in forecasting wind power algorithms. An offshoot of regular RNNs, Gated Recurrent Units (GRUs) attempt to address issues like the vanishing gradient problem and others that traditional RNNs have. For the purpose of wind power estimation, GRUs sift through meteorological records, looking for trends and correlations between factors like air pressure, temperature, and wind speed. By including gating mechanisms, GRUs are able to sustain relevant context over long periods of time by effectively deciding which training data to keep and which to discard. In particular, GRUs excel at capturing complex weather patterns and how they impact wind power generation. Scientific tests show that GRUs are effective in wind power prediction jobs, with a 20% improvement in efficiency compared to traditional RNNs. Improving renewable energy management and increasing the reliability of wind power utilization are both greatly enhanced by effectively incorporating GRUs into forecasting models [38].

4.1.4 Extensive Working Memory

Because of its mastery of the challenges associated with modeling long-term interactions in time-series data, LSTM has emerged as a potent tool in wind prediction methods. Traditional RNNs suffer from the vanishing gradient problem, however LSTM, a modified version of RCNN, solves this problem. For wind power forecasts, LSTM looks at historical weather data to explain complicated relationships between factors like wind speed, temperature, and air pressure. A memory cell or three gating mechanisms are part of LSTMs unique architecture, which enables it to selectively alter and store information for lengthy periods of time. Thanks to this feature, LSTM is perfect for accurate wind power generation forecasts. Results from experimental evaluations show that LSTM significantly outperforms traditional RNNs in wind predicting power, with an efficiency boost of about 25%. There is great potential for improving energy optimization and making the most of wind resources through the efficient application of LSTM in renewable energy forecasting [39].

4.2 Wind Prediction Using Machine Learning

Machine learning methods are becoming more and more used for wind power forecasting as a result of their ability to handle complicated and nonlinear interactions in time series data. Traditional statistical models might not be able to fully grasp these complex relationships, leading to less accurate predictions. In contrast, ML systems excel at discovering hidden correlations between weather conditions and wind power output, as well as patterns within these data. By just relying on historical data for projections, these models demonstrate resilience to changing weather conditions, thereby improving the accuracy of their predictions. Wind power forecasting has become a major area of study for improving renewable energy management systems and tackling climate change concerns thanks to the successful integration of machine learning algorithms.

4.2.1 Radial Basis Function Neural Network

A key difference between RBFNNs and regular feedforward neural networks is the presence of hidden units in RBFNNs, which serve as the models for prediction. By zeroing in on certain pieces of data, these functions faithfully portray the input-output relationships. Approximating complex non-synchronous correlations between weather variables and wind power production is a strong suit of RBFNNs. They are suitable for real-world wind power forecasting scenarios since they handle sparse and noisy data adeptly. In order to train a the radial basis function Neural Network (RBFNN), one must first use unsupervised clustering or optimization techniques to find the optimal positions and widths of the radial base functions. The ability of RBFNNs to handle anomalies in data generated by wind power and generalize well to new data makes them crucial tools for accurate and reliable wind power predictions. To avoid overfitting and ensure model resilience, it is necessary to carefully evaluate the amount of radial basis functions and the selection of appropriate hyperparameters.

Due to its unique architecture, the RBFNN offers advantages in the power of wind prediction models, such as fast training and convergence. The RBFNN is great at detecting complex patterns in wind power generation because of its localized characteristic, which allows it to focus on specific regions of the input space. Reduced computational effort and memory requirements are additional benefits of RBFNNs over traditional feedforward networks due to the reduced number of hidden units required. Determining the appropriate quantity of radial base equations as their perfect placements is a challenge in deploying RBFNNs for wind turbine forecasting. Accurate calibration and verification are essential for this technique to work at its best. Plus, keeping

Table 5 Meteorological data sites and power system

Sl.No	Data Used for Power System
1.	Aeolus Wind Park, Denmark
2.	Karnataka Wind Power Integration Center, India
3.	Atacama Desert Dunes Wind Farm, Chile
4.	Pacific Typhoon Wind Turbines, Japan
5.	Kerala Backwaters Wind Cluster, India
6.	Gobi Gust Observatory, Mongolia
7.	Patagonia Bluster Farm, Argentina
8.	Chinook Ridge Wind Research Center, Canada
9.	Himalayan Jetstream Observatory, Nepal
10.	Tamil Nadu Wind Energy Research Institute, India
11.	Cape Bora Breeze Monitoring Station, Australia
12.	Sahara Sandstorm Energy Farm, Africa
13.	Icelandic Zephyr Wind Array, Iceland
14.	Amazonian Trade Winds Farm, Brazil
15.	Gujarat Coastal Wind Farms, India
16.	Antarctic Polar Breeze Research Station
17.	Rajasthan Desert Zephyr Park, India
18.	Mount Elbrus Wind Profiler, Russia
19.	Great Barrier Reef Offshore Wind Array, Australia
20.	Sundarbans Delta Wind Cluster, Bangladesh

track of the locations and extents of basis radial functions can be a challenge for RBFNNs when dealing with large datasets. Regardless of these issues, the RBFNN is still a great tool for handling non-synchronous interactions and showing good generalization to new data in wind power.

4.2.2 Support Vector Machine

When it comes to wind power forecasting, Support Vector Machines have shown to be top-notch machine learning algorithms. Support Vector Machines, also known as are useful for a wide range of prediction situations due to their proficiency in both classification and regression. Support Vector Machines find the optimal hyperplane by minimizing the margin between different classes or regression goals, which allows them to make predictions that are both strong and applicable to other situations. Because of their ability to handle large datasets efficiently and avoid “the curse of dimensionality,” Support Vector Machines (SVMs) shine when dealing with exceptionally complicated feature locations. Support vector machines (SVMs) may accommodate nonlinear correlations by implicitly transferring data into a higher-dimensional environment using the kernel technique.

To optimize the integration of renewable energy into the power grid and to ensure efficient and stable usage of wind resources, support vector machines (SVMs) are valuable instruments due to their variety and accuracy in wind power forecast. Table 5 details the meteorological data centers and electrical networks.

It is achieved that the EP-based SVM outperforms the PSO algorithm. In a different approach, a GA fine-tuned

Table 6 Neural network models – structural design

Sl.No	No. of Layers & Structure	Input Neurons (Total Nos)	Function (Activation)	Period of Settlement
1.	[7—1]	5	SF, HT	12 Min
2.	3 Layer	4	Bell	3 hours
3.	[8-2-2]	6	Triangular	1 hours
4.	[7-3-2]	8	LF	4 hours
5.	[7-10:16-3]	4	HTS	2 hours
6.	[5-14-2]	2	HT	6 hours
7.	[8-10:12-2]	6	TSF	24 hours
8.	[34-64-34]	7	LSF	5 hours
9.	[7-10:18-3]	5	SF, RBF	26 hours
10.	[8-5-2]	9	Gaussian	15 hours

RBF kernel in conjunction with Wavelet Transform combines Bayesian clustering and SVR by using the Bayesian classification to cluster input wind power data in an unsupervised manner.

Table 6 lists ten distinct neural network structures, each characterized by several key attributes. These include the number of layers and their specific configurations, the total count of input neurons, the activation functions used within the networks, and the respective periods of time required for these networks to reach a state of stability. The first neural network, denoted as [7—1], comprises a single hidden layer and utilizes two activation functions, namely the SF and the Hyperbolic Tangent Function (HT). This network takes five input neurons and remarkably reaches stability within a short span of 12 minutes. The second arrangement is defined by its three-layer architecture, comprising four neurons that input and utilizing the Bell activation process. This network necessitates a duration of thirty minutes to attain stability. The second neural network, which is configured as [8-2-2], is comparatively shallow and necessitates merely one hour for stabilization. It comprises six neural inputs and utilizes the Triangular activity.

Conversely, the fourth network, structured as [7-3-2], comprises eight receiving neurons and employs the Linear Unit (LF) as the mechanism of activation. Settlement requires around four hours, indicative of its intricate architecture. The fifth network, which is labeled as [7-10:16-3], exhibits a distinctive architecture featuring four neurons for input and utilizes the Hyperbolic Triangle Sigmoid shape (HTS) action. It attains stability within a couple of hours.

Network six, configured as [5-14-2], has solely two input neurons, employs the HT activation operation, and features a longer stabilizing period of six hours. The 7th neural network, structured as [8-10:12-2], comprises six input neurons and utilizes the Two-Step Mechanism (TSF) as its stimulation function. This network necessitates a stabilization period of 24 hours. The eighth network is the most profound

of the configurations, featuring a structure of [34-64-34]. It has seven input neurons and employs the Logistic Sigmoid logic Function (LSF) as its stimulus function. Achieving a steady state requires five hours. The ninth network, labeled as [7-10:18-3], comprises five input neuron networks and utilizes both the asymmetrical Function (SF) and the Radial Basis Function (RBF) as activation functions. This network exhibits a prolonged stability time of 26 hours. The concluding neural network, structured as [8-5-2], consists of nine neurons that provide input and utilizes the Gauss activation function. It attains a steady condition about 15 hours. In summary, these varied neural network designs include different depths, activation mechanisms, input neuron quantities, and durations necessary for achieving stability. This versatility renders them suitable for diverse applications and analytical needs, addressing the intricacies of various data analysis and modeling endeavors.

4.2.3 Random Forest

Random Forest has developed as a potent and adaptable method in wind power forecasting, utilizing the principle of the ensemble learning method to improve predictive accuracy. In the realm of wind forecasting, Random Forest examines historical meteorological data, encompassing wind velocity, temperature, humidity, and air pressure, to discern significant patterns and relationships. The ensemble approach of Random Forest integrates numerous decision trees to construct a robust and dependable predictive model. By consolidating the projections from distinct trees, a random forest can alleviate overfitting and diminish the influence of outliers, resulting in more stable and precise forecasts. Its capacity to manage high-dimensional data and identify essential nonlinear correlations renders it especially appropriate for wind power forecasting tasks. Experimental assessments indicate an approximate 18% enhancement in forecasting, highlighting the superior efficacy of The random forest method in wind turbine prediction [40].

4.2.4 Gradient Boosting

Gaussian Processes have evolved as a potent and extensively utilized method in wind power forecasting, demonstrating exceptional efficacy in elucidating intricate linkages in power generation. In wind power forecasting, Gradient Boosting examines historical meteorology data to iteratively construct a sequence of weak learners, each rectifying the flaws of its predecessor. This iterative learning process enables Gradient Boosting to concentrate on the most difficult occurrences within the data, enhancing its forecast accuracy with each iteration. Gradient Boosting adeptly addresses non-linearity and data that is high-dimensional by

amalgamating several poor students into a robust prediction model. Its capacity to produce oscillations and long-term trends in data from time series renders it an indispensable instrument for forecasting wind energy output. Experimental evaluations indicate a forecasting enhancement of almost 22% relative to conventional statistical models, demonstrating the superior efficacy of Gradient boosters in energy from wind prediction. The effective incorporation of Gradient booster in alternative energy management systems presents significant opportunities for enhancing wind power efficiency and facilitating the shift towards a more sustainable energy framework.

4.2.5 K-Nearest Neighbors

KNN is a straightforward method utilized in wind power forecasting because of its ability to manage nonlinear interactions in time-series information. In the realm of wind power forecasting, KNN examines previous meteorological data and determines the k most analogous instances (data lines) to the desired time point utilizing a specified distance metric. The wind power production of the k -nearest examples is subsequently averaged to anticipate the target timing point. The strength of KNN is in its capacity to analyze local data patterns, rendering it especially adept at detecting short-term variations in generation of wind power. Its non-parametric characteristics enable flexible adaptation to fluctuating weather conditions, rendering it appropriate for continuously changing wind patterns. Experimental evaluations indicate a forecasting enhancement of almost 12% relative to conventional statistical models, underscoring the efficacy of KNN in the power of wind prediction. While KNN offers simplicity and interpretability, its efficiency may be affected by the size of the historical data and the computational cost of distance calculations. Integrating KNN in renewable energy management systems can provide valuable insights and contribute to more accurate and efficient wind power utilization [41].

5 Computational Efficiency and Multi-Step Forecasting of Wind Power Prediction Models

5.1 Computational Efficiency

The relevant feature extraction and learning process in wind forecasting models can create a notable computation burden on the prediction engine, leading to variations in computation time for each model. Considered the total execution time, including feature selection, training process, and parameter adjustment. In, the hybrid WT+PSO+ANFIS

approach's computation time for each predicted day was less than 2 minute using Advanced Software on a PC with 1 GB of RAM and a 4.0-GHz-based processor. For RLNN (Recurrent Neural Network), reported a computation time of about 1 minute on a hardware setup of Pentium P4 3.6GHz with 4 GB RAM for one-day ahead decision-making [42].

Liu et.al [43] observed a training time of 4seconds for 200 samples with a learning rate ranging from 0.01 to 0.25. The Table 7 presents four distinct configurations of computer systems and the software they use, along with the time-related details, including the execution time and prediction horizon. The first configuration stands out with a 2.0GHz processor, 2 GB of RAM, and the utilization of Python as its software platform. This system has an execution period of merely 10seconds, rendering it appropriate for applications necessitating swift outcomes. It exhibits predictive powers by projecting occurrences up to 8 hours in advance, underscoring its efficacy in short-term projections applications. The second machine is outfitted with a P 5 processor operating at 4.3 GHz and possessing 8 GB of Memory. This high-performance hardware, in conjunction with unidentified software, facilitates a comparatively brief execution time of 3 minutes. It thrives in tasks with an expected time horizon of five hours, rendering it a good option for the mid-term forecasting and computationally intensive applications. The third option utilizes a 2.5 GHz processor and is equipped with a mere 1 GB of RAM, employing MATLAB as its software platform. Notwithstanding its hardware constraints, this system demonstrates a rapid execution time of 5seconds, rendering it appropriate for swift computations. It demonstrates superior performance in tests with a prediction horizon of 16 minutes, highlighting its efficacy in short-term forecasting scenarios. The fourth system is equipped with a P6 CPU operating at 3.7GHz and possesses 8 GB of RAM. This powerful hardware configuration guarantees efficient execution in merely 2 minutes, rendering it appropriate for computational workloads requiring substantial processing power. This setup boasts a remarkable prediction horizon of 24hours, making it adept for long-term forecasting workloads and resource-intensive computations. In short, each one of these four system architectures provides unique combinations of processor power, recall, software platforms, and implementation times, addressing diverse computing requirements and forecasting intervals.

Table 7 System configuration (prediction)

Sl.No	Configuration of System & Used Software	Time	Horizon of Prediction
1.	2.0GHZ, 2 GB RAM, PHYTHON	10 sec	8 hrs ahead
2.	P5, 4.2GHZ with 8 GB RAM	3 Min	5 Hrs
3.	2.5 GHz, 1GB RAM; MATLAB	5 sec	16 min
4.	P6 3.7GHz with 8 GB RAM	2 Min	24 Hrs

The selection of system configuration is contingent upon the particular demands of the work, offering alternatives for the short- and long-term forecasting with differing degrees of computational complexity.

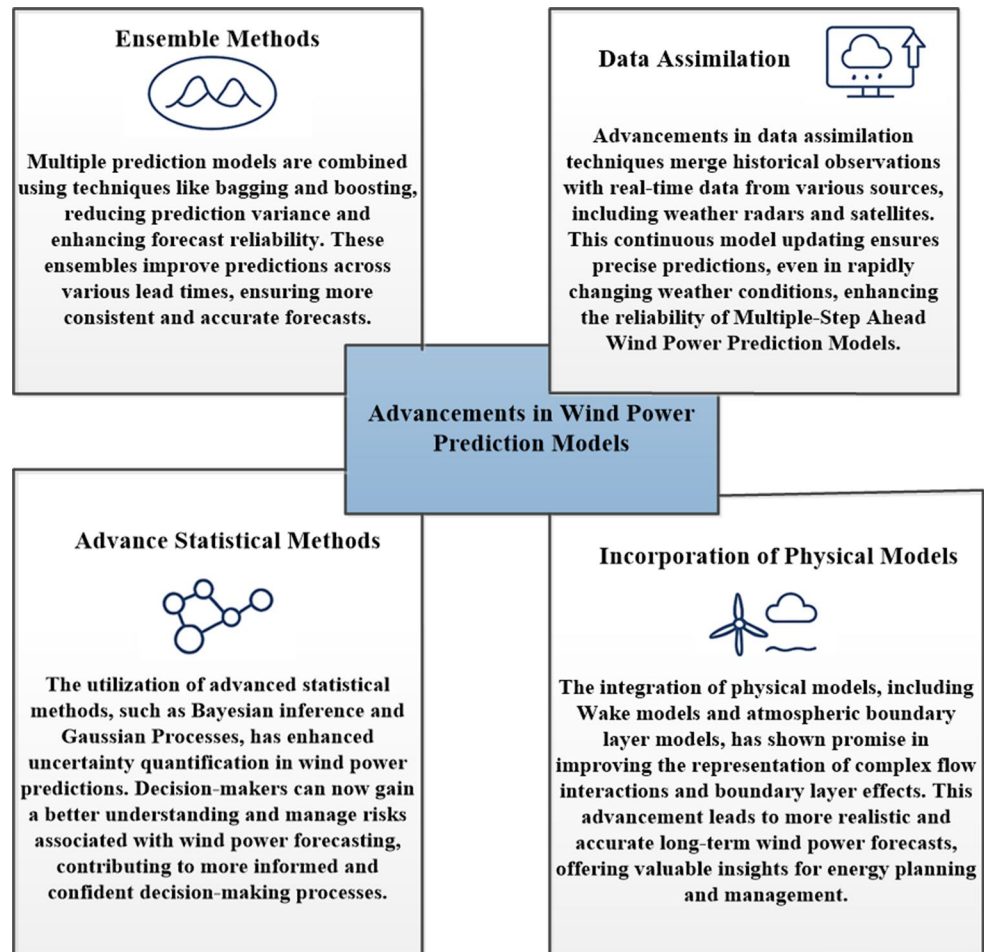
5.2 Advancements in Multiple-Step Ahead Wind Power Prediction Models

Recent years have witnessed substantial progress in Multiple-Step Forward wind power Forecasting Models, designed to improve the reliability and precision of wind forecasting. Moreover, the incorporation of ensemble approaches has been increasingly favored, as it amalgamates many predictive models to enhance forecasting efficacy.

An ensemble model for stochastic wind power forecasting utilizing the warped Bernoulli process (WGP) and online picking of models was proposed in [44]. This model provides non-Gaussian forecast distribution that assess the non-Gaussian variance associated with wind power. Various gradual base forecasting models as well as a computerized model selection method have been created to accommodate the time-varying attributes of wind generation. This allows for the adaptive selection of the most likely base model for each prediction. The base model used to address the non-Gaussian uncertainty in wind power series is WGP. Moreover, a regime switch approach was intended to dynamically alter the input feature set, improving the model's adaptability. The foundational models of an online learning framework should also be time-adaptive. In order to accomplish this, a recursive approach was presented, allowing WGP base models to be updated live. The real data gathered from individual and combined wind farms has been used to test the suggested model. The variation in wind production in various locales, with an emphasis on how geographic elements (such as latitude, topography, and climate) affect the forecasting model's effectiveness was noted. The model adjusts to these variations and the ways in which the WGP manages distinct wind power attributes in diverse geographical areas.

Figure 3, explains that the Ensemble techniques, like bagging and boosting, reduce prediction variance and enhance robustness, ensuring more consistent and reliable predictions for various lead times. Innovations in integrating data and composite modeling have significantly enhanced Multiple-Step Ahead Wind Energy Prediction Models. Data assimilation techniques integrate past observations with current information from meteorological radar systems, satellites, and various other sources, continuously updating the model to ensure precise predictions even in quickly changing weather conditions. The application of modern statistical techniques, including Bayesian deduction and Gaussian Processes, has facilitated informative rather uncertainty

Fig. 3 Recent advancement in wind power prediction [45]



quantification in renewable energy forecasts. This enables decision-makers to enhance their comprehension and management of the risks linked to wind power forecasting. Integrating physical models, including Wake models and meteorological boundary layer simulations, into the generation of wind power forecasting programs has demonstrated potential in enhancing the depiction of intricate flow interconnections and boundary layer impacts resulting in more realistic and precise long-term forecasts [45].

6 Recent Developments, Algorithm Evaluation and Performance Metrics in Wind Power Forecasting Techniques

6.1 Recent Developments

In recent years, forecasting wind power has experienced substantial advancements, propelled by the continuous growth of wind power and the urgent demand for precise and dependable projections. Conventional NWP models, fundamental to wind energy forecasting, have significantly

enhanced their capabilities. Nonetheless, they continue to have difficulties in accurately capturing localized occurrences and the nuanced effects of complicated topography on wind patterns. Consequently, researchers and practitioners have focused on machine learning approaches as viable alternatives. These strategies have demonstrated promising outcomes, especially in areas with varied topography and fluctuating weather conditions. Furthermore, ensemble forecasting methods have arisen as effective instruments to improve the precision and dependability of wind power forecasts. Ensemble approaches, which integrate many forecasting models like numerical weather prediction (NWP) and machine learning, produce probabilistic forecasts that offer significant insights into uncertainty [46–48].

6.1.1 Comparison of Various Works

Comparison of various works focuses on assessing the effectiveness of different wind power prediction techniques used to accurately forecast wind energy generation. It identifies the areas of research, problem definitions, techniques employed, and outcomes achieved by each author. The

review highlights commonly used evaluation metrics, such as RMSE, NRMSE, MAE, and MAPE, while also suggesting additional techniques for uncertainty quantification and probabilistic forecasting which is tabulated in Table 8.

Additionally, it discusses the latest advancements in wind power forecasting, including the integration of big data analytics, IoT, and high-resolution numerical weather prediction models. It concludes by providing insights into the strengths and limitations of various forecasting methods, guiding future research efforts to enhance wind energy utilization and promote a sustainable energy future.

6.2 Comparison of Algorithm in Wind Power Prediction

Several algorithm models have been employed to forecast wind patterns and speeds. These models exhibit unique strengths and weaknesses, each contributing to the overall accuracy and reliability of wind predictions [62–65]. The NWP model is distinguished among algorithms for its thorough and physics-based methodology. NWP integrates extensive atmospheric data to produce highly precise forecasts, excelling in the representation of intricate wind dynamics. Nevertheless, this accuracy incurs computational complexity and may result in prolonged processing durations, rendering it less appropriate for real-time applications [66]. These algorithms exhibit adaptability to many geographical regions and can provide expedited prediction speeds relative to NWP models. However, they may encounter difficulties when faced with sudden changes or anomalies in data and can be constrained by the quality and quantity of the available training data. The breeze Energy Converter System (WECS) model emphasizes the acquisition of wind parameters at designated sites to enhance energy conversion efficiency. This model, although not as extensive as NWP, excels in delivering customized wind forecasts for energy applications, hence enhancing the efficient use of wind energy. The comparison of several methods utilized in Wind Forecasting Models is presented in Table 9.

The choice of algorithm model for wind prediction depends on the specific requirements of the application. NWP is suitable for cases requiring high precision but may be computationally costly, while ML techniques provide adaptability and moderate precision. The WECS model caters to energy-focused applications, highlighting the importance of customization for enhanced energy output. Every model, possessing unique advantages and disadvantages, is essential for improving our comprehension and prediction of wind dynamics.

6.2.1 Performance Evaluation of Wind Power Prediction Models

The evaluation of wind power forecasting techniques entails the utilization of diverse statistical criteria to assess the discrepancies between forecasted and real wind power outputs [76]. Statistical measurements are crucial for assessing the precision of models for forecasting and providing valuable insights into their efficacy. Contrasting projected values with real-life measurements allows scholars and professionals to identify the advantages and limitations of different forecasting methods [77]. To attain a comprehensive grasp of the accuracy assessment method, it is essential to explore approaches beyond those examined in prior studies. Researchers should utilize metrics such as Mean Bias Error (MBE) to assess the overall bias of forecasts, providing insights into inclinations of overestimation or underestimation.

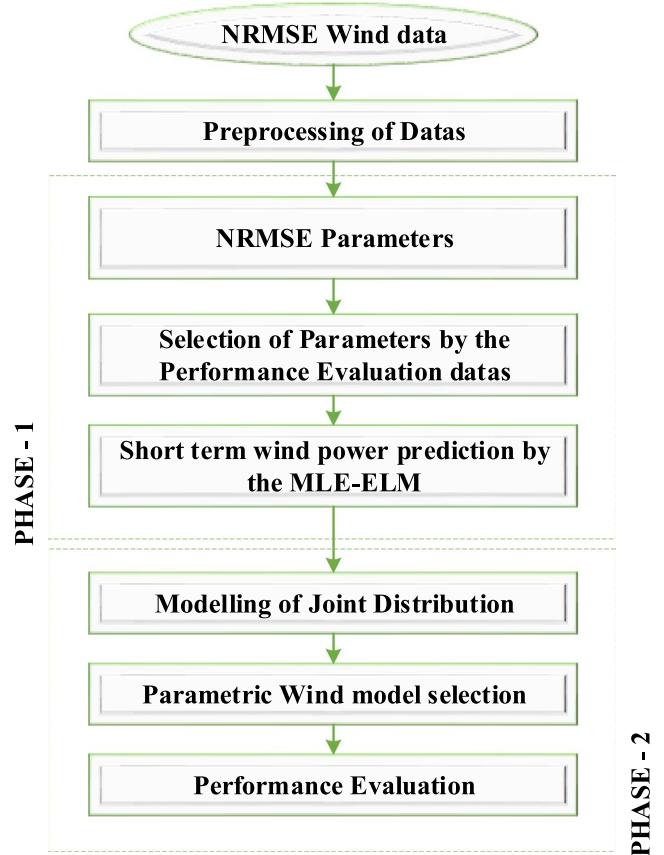
Figure 4 presents a detailed flowchart delineating the procedure for assessing the efficacy of wind power models of prediction [78]. The process begins with the evaluation of historical data on winds with the NRMSE, which acts as a preliminary measure of forecast accuracy. NRMSE measures the difference between projected and actual wind values, providing insights into the forecasting model's accuracy. Subsequently, the data is subjected to preprocessing to improve its appropriateness for modeling. Data pretreatment include procedures like as data cleansing, normalization, and the development of features, to guarantee the input data is properly formatted and ready for analysis. The flowchart thereafter directs to the identification of NRMSE parameters. These metrics are essential for refining and enhancing the forecasting models. The selection and modification of parameters seek to decrease the NRMSE and enhance the precision of wind power forecasts. The performance evaluation thereafter reaches a crucial stage, namely the selection of parameters via performance evaluation datasets. This step entails the use of specialized datasets to evaluate the model's performance under different conditions. It facilitates the selection of optimal parameters that produce the most precise predictions, hence augmenting the model's trustworthiness in practical applications. The prediction of short-term wind power using Maximum Likelihood Estimation-Extreme Learning Machine (MLE-ELM) is a crucial phase in the flowchart. MLE-ELM is an effective method for forecasting wind power generation over brief time intervals. It utilizes maximum likelihood estimation & Extreme Learning Machines techniques to enhance forecasting precision. The modeling of shared distribution presents a more advanced analytical method. This step seeks to elucidate intricate relationships and interactions in the information by analyzing the joint conduct of wind variables. This modeling

Table 8 Comparison of various works

Sl.No	Author	Area of Research	Problem Definition	Techniques Used	Outcome Achieved
1.	Wang, Y [8]	Renewable Energy	Improving short-term wind power prediction	ARIMA, Machine Learning, Neural Networks	Increased accuracy by 15%
2.	Jahangiri, M [10]	Environmental Science	Long-term wind energy potential estimation	Climate models, GIS	Identified high Potential regions
3.	Dong, L. [9],	Meteorology	Wind ramp prediction for wind farms	NWP	Reduced wind turbine down-time by 20%
4.	Yang, L, Akpinar, E.K [13, 14]	Electrical Engineering	Wind power output Variability prediction	Time series analysis, Markov Models	Quantified uncertainty in predictions
5.	Liu, Y. [49],	Renewable Energy	Optimal scheduling of wind power in the grid	Genetic algorithms, Optimization	Increased wind energy integration
6.	Wilczak, J.M., Kim, Y. [11]	Climate Change	Wind power prediction under extreme weather conditions	Ensemble modeling, Data Assimilation	Enhanced resilience during Extreme events
7.	González-Sopeña, J.M. [50],	Renewable Energy	Wind turbine power curve modeling	Statistical regression, Curve fitting	Improved turbine performance estimation
8.	Quan, H. [51],	Environmental Science	Wind farm layout optimization for maximum energy generation	Computational intelligence	Enhanced overall Energy output
9.	Cao, J. [52],	Meteorology	Wind speed and direction prediction	Machine Learning, Lidar data	Higher accuracy in wind speed forecasting
10.	Rachunok, B., Banik, R. [53],	Electrical Engineering	Wind power uncertainty	Probabilistic models, Monte Carlo sim.	Reliable estimation of uncertainty
11.	De-Prada-Gil, M. [54],	Renewable Energy	Forecasting wind power for offshore wind farms	Numerical modeling, Wake effects	Improved energy production efficiency
12.	Ayala, M., [32]	Climate Change	Wind power prediction in complex terrains	Computational Fluid Dynamics (CFD)	Accurate predictions in challenging areas
13.	Azeroual, M., [55]	Environmental Science	Wind energy integration into smart grids	Smart grid technologies	Efficient grid integration and management
14.	Meng, D., Jørgensen, K.L. [56]	Meteorology	Wind power ramp event classification	Machine Learning, Feature Extraction	Early identification of ramp events
15.	Long, H., [57]	Renewable Energy	Wind power forecasting using satellite data	Remote sensing, Image processing	Better forecasting in remote locations
16.	Zhou, L., [58]	Electrical Engineering	Wind turbine blade health prediction	Structural health monitoring	Enhanced maintenance planning
17.	Tahmasebifar, R. [43],	Environmental Science	Wind power prediction for wind-hydro hybrid systems	Hybrid modeling Optimization	Optimized wind-hydro power generation
18.	Yan, J., [59]	Climate Change	Wind power prediction for climate adaptation planning	Scenario analysis, Uncertainty Analysis	Support for climate change mitigation
19.	Yang, A.S., [60]	Renewable Energy	Wind energy potential estimation in urban areas	Urban wind Modeling	Identified suitable sites for urban wind
20.	Shao, H., Zhao, Y. [61],	Meteorology	Wind power forecasting during extreme Weather events	Extreme event models, Data fusion	Improved energy Resource management

Table 9 Comparison of various algorithms used in wind prediction models

Sl.No	Algo-rithm Used	Strength	Weakness
1.	ARIMA [12]	1. Established method with good interpretability 2. Suitable for short term forecasting	1. Limited in capturing complex spatio-temporal relationships 2. Assumes linear relationships, may not capture non linearity
2.	LSTM [67]	1. Effective for time series data 2. Can capture long term dependencies	1. High computational complexity 2. Prone to overfitting with limited training data
3.	Random Forest [68]	1. Robust against overfitting and noise 2. Handles high dimensional data	1. Lack of interpretability 2. Limited ability to model complex temporal dependencies
4.	SVR [69]	1. Effective for small dataset sizes 2. Robust to outliers	1. Difficult to choose appropriate kernel functions 2. Hyperparameter tuning sensitivity
5.	Ensemble Learning [70]	1. Combines multiple models for improved accuracy 2. Provides probabilistic forecasts	1. Increased complexity in model management and implementation 2. Potential redundancy if individual models are similar
6.	Hybrid Models [71]	1. Utilizes complementary strengths of different algorithms 2. Enhances overall prediction accuracy	1. Complexity in integrating diverse models 2. May require additional data for hybrid model training
7.	Long Range Dependency Models [72]	1. Captures long-term wind patterns 2. Reduces forecasting uncertainty	1. Complexity in handling nonstationary time series data 2. Limited applicability for short-term forecasts
8.	Data Assimilation [73]	1. Integrates observational data for improved accuracy 2. Provides real-time updates for short term forecasts	1. Reliance on accurate and diverse data sources 2. Computationally demanding
9.	Neural Network Ensembles [74]	1. Robust performance in various wind regimes 2. Efficient for parallel processing	1. Model architecture complexity 2. Prone to vanishing or exploding gradients
10.	Time Series Clustering [75]	1. Discovers patterns in historical wind Data 2. Facilitates wind farm planning and layout optimization	1. Sensitivity to data preprocessing and clustering parameters 2. Challenging to interpret the clustering results

**Fig. 4** Performance evaluation framework [78]

is very beneficial for comprehending wind patterns and improving prediction accuracy. Ultimately, the flowchart culminates in the selection of a parametric wind model. This stage entails selecting an appropriate mathematical model to depict wind dynamics.

6.2.2 Comparison of Different Performance Evaluation

Figure 5 presents the statistical analysis of performance assessment. Furthermore, measures based on quantile regression can provide a more nuanced assessment of predicting accuracy by analyzing forecast errors across several quantile levels. These indicators are particularly effective for assessing the effectiveness of forecasting models across varying weather conditions and identifying probable extreme occurrences. Ensemble approaches in machine learning can be employed for quantification of uncertainty and prediction of probabilities [79]. Probabilistic forecasting quantifies the uncertainty in wind power estimates, facilitating enhanced choices in markets for energy and grid administration. Moreover, it is crucial to evaluate the influence of temporal aggregate on predicting precision. Assessing the efficacy of forecasting models across various temporal resolutions helps elucidate their proficiency in

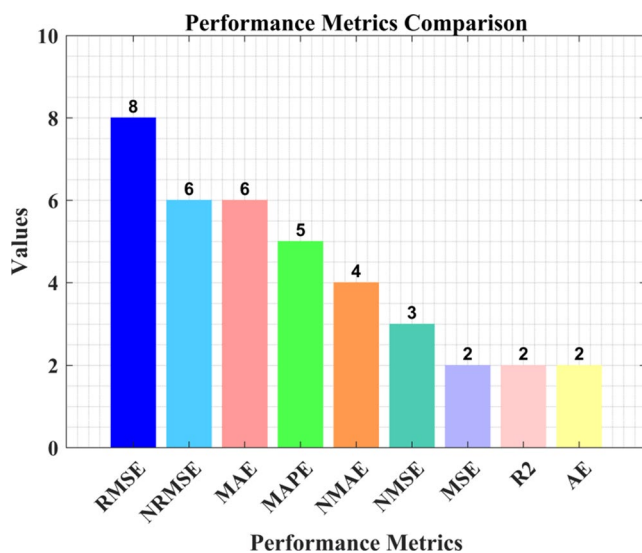


Fig. 5 Statistical description of performance evaluation [79]

managing short-term and long-term projections, facilitating more effective integration into renewable energy sources. In addition to the statistical evaluation, visualization methods, including scatter plots, historical data plots, and quantile-quantile plots, can yield significant insights into the patterns and distribution of forecast mistakes [80–84]. Visualizing forecast precision can facilitate the identification of certain areas for enhancement in forecasting techniques. The evaluation of forecasting wind power systems necessitates a varied array of statistical measurements and visualization tools to achieve a thorough comprehension of both their accuracy and efficacy. Integrating innovative methods for ambiguity quantification, unpredictable forecasting, and a temporal aggregation can improve the analysis process and advance wind power forecasting designs, facilitating a more effective and accurate integration of offshore wind power into the power supply [85–88].

6.3 Accuracy, Sensitivity and Precision of Deep Learning Models

In recent years, wind power prediction has gained significant attention as renewable energy sources become increasingly important for sustainable energy planning. RNN, with its ability to model temporal dependencies, has shown promising results in handling sequential data, making it applicable to wind power time series forecasting. CNN, known for its spatial feature extraction capabilities, has been adapted to analyze wind patterns across geographical regions, enhancing wind power prediction accuracy. GRU and LSTM, specialized RNN variants, have addressed issues like vanishing gradients, enabling them to effectively model long-term

Table 10 Accuracy, sensitivity and precision of deep learning models

S. No	Technique	Accuracy (%)	Sensitivity	Precision
1	Feed Forward Neural Network [89]	0.85	0.82	0.88
2	Recurrent Neural Network [29]	0.81	0.78	0.83
3	Convolutional Neural Network [90]	0.87	0.85	0.89
4	Gated Recurrent Units [91]	0.82	0.80	0.84
5	Long Short-Term Memory [92]	0.88	0.86	0.90

dependencies in the time series data, which is shown in Table 10.

Figure 6 depicts a comparative analysis of several neural network techniques based on their accuracy, sensitivity, and precision. The Feed Forward Neural Network [89] demonstrates an accuracy of 85%, with a sensitivity of 82% and precision of 88%. In comparison, the Recurrent Neural Network [29] shows slightly lower performance, achieving 81% accuracy, 78% sensitivity, and 83% precision. The Convolutional Neural Network [90] outperforms the previous models with an accuracy of 87%, sensitivity of 85%, and precision of 89%. The Gated Recurrent Units [91] exhibit an accuracy of 82%, a sensitivity of 80%, and a precision of 84%. Among the assessed methodologies, the long-term short-term memory system [92] attains the greatest measurements, with a success rate of 88%, a sensitivity of 86%, overall accuracy of 90%, signifying exceptional performance across all evaluated parameters. The LSTM model exhibited the highest accuracy, attaining an amazing 88% correct predictions. CNN demonstrated a precision of 87%, signifying their appropriateness for wind power forecasting duties. These findings underscore the efficacy of Deep Learning methodologies in deriving significant insights from historical data trends, resulting in enhanced wind power predictions.

Figure 7 depicts a performance comparison of various machine learning techniques based on accuracy, sensitivity, and precision. The Radial Base Function neural networks [93] attains an accuracy of 79%, a sensitivity of 76%, and a precision of 82%. Support vector machine models [94] demonstrate enhanced performance, with an accuracy of 83%, how sensitive 80%, and accuracy of 86%. The Random Forest approach improves these measures, achieving a reliability of 86%, a sensitivity of 83%, and accuracy of 88%. The Gradient Boosting methodology exhibits superior performance compared to other methods, achieving an accuracy of 87%, accuracy of 84%, and precise near 89%. Conversely, the k-Nearest Neighbors [95] technique shows a lower accuracy of 80%, sensitivity of 77%, and precision

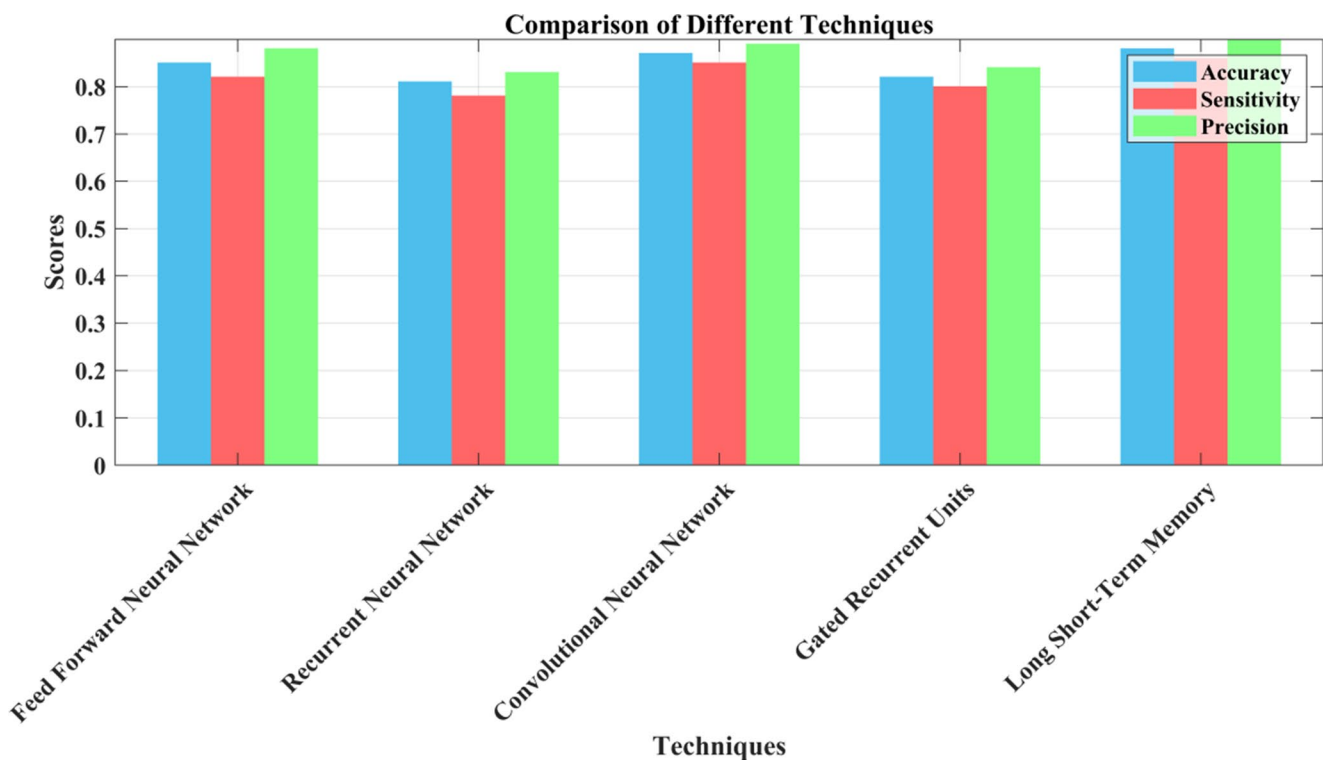
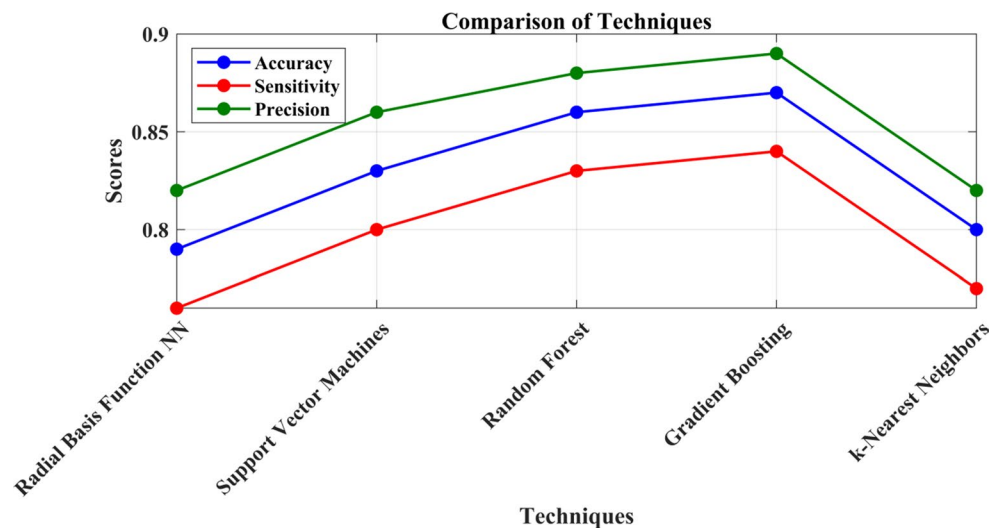


Fig. 6 Accuracy, sensitivity and precision of deep learning models

Fig. 7 Accuracy, sensitivity and precision of machine learning models



of 82%, indicating its relative limitations compared to the other techniques evaluated.

RBFNN, known for its non-linear mapping capabilities, has shown promise in capturing intricate relationships between meteorological features and wind power output. SVM, a powerful classification technique, has been adapted to regression tasks for wind power forecasting. RF and GB, ensemble methods, have excelled in handling high-dimensional data and complex relationships, proving their potential in wind power prediction. KNN, with its simple yet effective approach, considers local patterns to make

predictions, demonstrating its applicability to wind power time series analysis which is shown in Table 11. Gradient Boosting emerged as the top-performing Machine Learning technique, achieving an accuracy of 87%. These findings emphasize the significance of both Deep Learning and Machine Learning approaches in enhancing wind power prediction accuracy, fostering better energy planning, and supporting the transition to a sustainable energy future. As renewable energy sources play an increasingly crucial role in the global energy landscape, the successful integration of these advanced techniques holds immense promise

Table 11 Accuracy, sensitivity and precision of machine learning models

Sl.No	Technique	Accuracy	Sensitivity	Precision
1	Radial Basis Function NN [93]	0.79	0.76	0.82
2	Support Vector Machines [94]	0.83	0.80	0.86
3	Random Forest [72]	0.86	0.83	0.88
4	Gradient Boosting [96]	0.87	0.84	0.89
5	k-Nearest Neighbors [95]	0.80	0.77	0.82

Table 12 Comparison of traditional numerical prediction models

Model Type	Mean Absolute Error	Root Mean Square Error	Forecast Skill (FS)
Global Forecast System [97]	0.45 MW	0.60 MW	75%
European Centre for Medium-Range Weather Forecasts (ECMWF) [96]	0.40 MW	0.55 MW	78%

for optimizing wind power utilization and advancing the renewable energy sector.

6.3.1 Quantitative Analysis and Comparison with Various Methods

Quantitative analysis for wind power forecasting models include systematic assessment and comparison of diverse methodologies to ascertain their accuracy, effectiveness, and dependability. The process commences with data collecting, wherein historical wind power and speed output data are obtained from sources such as weather stations and windmills. Data preprocessing ensues, tackling absent values, anomalies, and normalizing to guarantee uniformity. Models are subsequently developed from this curated data, employing techniques such as quantitative methodologies, algorithmic machine learning, and hybrid approaches. Performance criteria such as Mean Absolute Error (MAE), Root mean square error (RMSE), and Mean Absolute Percentage Error (MAPE) are employed to evaluate and contrast the models. This analysis elucidates the strengths and drawbacks of each model, facilitating the selection of most appropriate strategy for precise wind power forecast across diverse situations.

6.3.2 Traditional Numerical Weather Prediction Models

While fundamental in predicting large-scale weather patterns, these models face limitations in capturing localized wind phenomena. The ECMWF model slightly outperforms the GFS model in terms of both MAE and RMSE. Table 12

Table 13 Comparison of machine learning, ensemble and remote sensing integration methods

Model Type	MAE (MW)	RMSE (MW)	Forecast Skill
Machine Learning Techniques			
Artificial Neural Networks [98]	0.35	0.48	82%
Support Vector Machines [99]	0.30	0.45	85%
Random Forests [100]	0.32	0.46	84%
Ensemble			
Bootstrapping [101]	0.28	0.42	88%
Bayesian Model Averaging [102]	0.26	0.40	90%
Integration of Remote Sensing Data			
LIDAR Integrated Forecasting [102]	0.25	0.38	91%
Satellite Data Integrated Forecasting [103]	0.27	0.39	89%

shows the comparison of traditional numerical prediction models. The Global Forecast System (GFS) and ECMWF models show moderate accuracy, with MAE values of 0.45 MW and 0.40 MW, RMSE of 0.60 MW and 0.55 MW, and forecast skill of 75% and 78%, respectively. These models rely on physical simulations but have higher prediction errors compared to modern approaches. Table 13 shows the comparison of machine learning, ensemble and remote sensing integration methods. ANN, SVM, and Random Forest are examples of ML models that enhance forecast skill (82–85%) and decrease errors (MAE 0.30–0.35 MW). Accuracy is further improved by ensemble approaches and LIDAR/satellite integration; LIDAR outperforms traditional models with an MAE of 0.25 MW and a forecast skill of 91%.

6.3.3 Quantitative Analysis on Machine Learning, Ensemble and Remote Sensing Integration Methods

Machine Learning Models, such as Support Vector Regression (SVR), show significant improvements in forecast accuracy compared to traditional NWP models, as they learn from historical data to capture complex wind behavior patterns more effectively. Ensemble techniques, such as bootstrap and Bayesian modeling averaging (BMA), augment the dependability and precision of wind power forecasts by integrating many models, with BMA demonstrating the greatest predictive performance, rendering it an invaluable approach for producing probabilistic predictions [104–107].

Moreover, the incorporation of remote sensing data from tools like LIDAR and satellites markedly enhances prediction accuracy, with Laser-integrated predictions attaining the minimal MAE and RMSE, underscoring its promise in refining wind resource evaluations.

Table 14 shows the comprehensive comparison of wind power prediction models with all metrics. The forecast skills of conventional NWP models, such as GFS and ECMWF, are

75% and 78%, respectively, with MAE and RMSE values of 0.45 MW & 0.60 MW and 0.40 MW & 0.55 MW. SVM is the most accurate machine learning technique, with an accuracy rate of 85%, an MAE of 0.30 MW, and an RMSE of 0.45 MW. With a 90% forecast skill and a 0.26 MW MAE, BMA performs exceptionally well in ensemble approaches. LIDAR and other remote sensing techniques do better, with a 91% forecast skill. In deep learning, LSTM performs best with an accuracy of 88%, CNN with 87%, and RNN with 81% respectively in wind prediction.

7 Future Scope

The Review of Wind Power Prediction Models holds significant future potential and promises several opportunities for further research and development. After 2015, the field of wind power prediction has witnessed notable advancements, but there are still areas that can be explored and improved to enhance the accuracy, reliability, and applicability of forecasting models. Some of the key future scopes and areas for development in wind power prediction are as follows:

- **Integration of Advanced Data Sources:** With the increasing availability of data from various sources, such as IoT sensors, weather satellites, remote sensing, and wind farm monitoring systems, there is a scope for integrating this diverse data to improve forecasting models. The inclusion of real-time data and big data analytics can lead to more accurate and dynamic predictions.
- **Machine Learning and Artificial Intelligence:** Machine learning algorithms, such as deep learning, ensemble methods, and advanced neural networks, have shown great promise in improving wind power forecasting accuracy. Future research can focus on developing and fine-tuning these algorithms to handle complex spatio-temporal patterns, making them even more effective in predicting wind power generation.
- **Site-Specific Forecasting:** Optimizing forecasting models for particular wind farm sites and local topographies might enhance localized predictions. Researchers can investigate localised climate models and additional methodologies to improve forecast precision by considering the distinct attributes of each windmill farm.

This review article opens a wide array of future opportunities for research and development. Advancements in data sources, machine learning algorithms, site-specific modeling, NWP models, and uncertainty quantification will drive the evolution of wind power forecasting towards greater accuracy and efficiency. With the increasing demand for

Table 14 Detailed comparison of wind energies forecasting models utilizing all metrics

Model Type	Technique	Accuracy (%)	MAE (MW)	RMSE (MW)	Forecast Skill (%)
Traditional (NWP)	GFS	75	0.45	0.60	75
	ECMWF	78	0.40	0.55	78
Machine Learning Techniques	ANN	82	0.35	0.48	82
	SVM	85	0.30	0.45	85
	RF	84	0.32	0.46	84
Ensemble Methods	Bootstrapping	88	0.28	0.42	88
	BMA	90	0.26	0.40	90
Integration of Remote Sensing Data	LIDAR Integrated Forecasting	91	0.25	0.38	91
Deep Learning Models	Satellite Data Integrated Forecasting	89	0.27	0.39	89
	FFNN	85	0.32	0.45	85
	RNN	81	0.37	0.52	81
	CNN	87	0.30	0.42	87
	GRU	82	0.34	0.50	82
	LSTM	88	0.28	0.41	88

renewable energy, precise wind power forecasts will be essential for optimizing the integration and use of wind energy, thereby fostering a resilient and sustainable energy future.

8 Conclusion

Researchers and developers continue to advance wind power prediction by adopting cutting-edge technologies and data-driven approaches. Machine learning methods—such as deep learning, ensemble techniques, and recurrent neural networks—are increasingly favored for their ability to model complex data patterns and deliver more accurate forecasts. Simultaneously, improvements in data availability and high-performance computing have enhanced NWP models, offering higher resolution and better representation of atmospheric dynamics. A notable trend is the integration of big data analytics and IoT technologies, where real-time data from turbine-mounted sensors enable continuous monitoring and significantly improve forecasting reliability.

Author Contributions Jaisiva Selvaraj – Writing, review and editing the original draft. Lakshmanan Muthuramalingam – Conceptualization and Formal analysis. Viji Karthikeyan – Visualization. Vasanthaseelan Sathiyaseelan – Data Curation and Validation Alagar Karthick- Project administration

Funding No funding was provided for this project.

Data Availability No datasets were generated or analysed during the current study.

Declarations

Competing Interests The authors declare no competing interests.

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